1. Introduction

1.1. Project Background

Introduce the general topic of automatic image captioning and explain its purpose: generating natural-language descriptions of images.

Present the main idea of the project, which is to study how different Retrieval-Augmented Generation (RAG) techniques influence the quality, relevance, and visual grounding of generated image captions.

Briefly explain that RAG combines information retrieval with text generation. Instead of relying exclusively on the information learned by a model, a RAG system retrieves relevant information from a knowledge base to support the generation process.

1.2. Importance of Data Exploration and Pre-processing

Explain why data quality is essential for a reliable comparison of RAG techniques.

Discuss the importance of:

- Understanding the structure and distribution of the dataset.
- Detecting missing, corrupted, or duplicated data.
- Exploring image annotations and reference captions.
- Standardizing images and preparing textual data.
- Creating reliable dataset splits.
- Preventing information leakage between the knowledge base and evaluation data.

1.3. Objectives of Phase 1

Describe the objectives of the current phase:

- Explore the MS COCO 2017 dataset.
- Analyze object categories, scene complexity, and caption characteristics.
- Validate the integrity of images and annotations.
- Preprocess images and captions.
- Establish reproducible knowledge-base, validation, and test splits.
- Organize metadata and prepare the data for future embedding generation and retrieval experiments.

Conclude by explaining that this phase establishes the common data foundation for the subsequent implementation and comparison of RAG techniques.

2. Dataset Description

2.1. Overview of the MS COCO Dataset

Introduce MS COCO 2017 as the dataset used in the project.

Explain that it contains real-world images representing different objects, scenes, and interactions. The dataset provides image annotations, object categories, and naturallanguage captions.

Clarify that the project uses two annotation types for different purposes:

- Caption annotations, which provide natural-language descriptions of images.
- Instance annotations, which provide object categories, bounding boxes, and related information.

2.2. Dataset Sources and Files

Describe the principal annotation files used in the notebooks:

- captions_train2017.json: reference captions for training images.
- instances_train2017.json: object-instance annotations for training images.
- captions_val2017.json: reference captions for test images selected from the validation split.
- instances_val2017.json: object-instance annotations available for the validation images, if required by the analysis.

Explain that the images are downloaded separately and associated with their annotations through image IDs.

2.3. Dataset Organization

Present the repository structure and explain the purpose of its main directories.

- data/annotations/: original COCO annotation files.
- data/images/train/: downloaded images used for the knowledge base and validation subset.
- data/images/test/: images reserved for final evaluation.
- data/processed/images/: images after orientation correction and RGB conversion.
- data/processed/images_resized/: resized images.
- data/processed/captions/: processed captions and caption audit information.

- data/processed/exploration/: statistical tables generated by the exploration notebook.
- data/processed/test/: test reference captions.
- splits/splits_all.json: image IDs defining the dataset splits.

Mention that some of these files and folders are generated when the notebooks are executed.

2.4. Dataset Splits and Their Roles

Describe the split configuration documented in the project README:

Split	Number of images	Role
Knowledge Base (KB)	10,000	Retrieval corpus
Validation	100	Hyperparameter tuning and validation
Test	100	Final evaluation

Explain that the validation subset is selected from the knowledge base according to the project configuration, whereas the test images are selected from COCO val2017 and must remain independent of the knowledge base.

The project uses seed 42 to support reproducibility.

The test set is also designed to include different types of scenes:

- Simple scenes containing 1–3 objects.
- Complex scenes containing at least 10 objects.
- Multi-category scenes containing at least 5 object categories.
- Interaction scenes containing a person and other objects. Scenes representing varied contexts.

Explain that these groups help assess the behavior of the future captioning systems under different visual conditions.

3. Data Exploration

3.1. Dataset Structure and Integrity

3.1.1. Dataset Loading and Annotation Inspection

Explain how the COCO API (pycocotools) is used to load image IDs, captions, and object annotations.

Describe the information extracted from the annotations:

- Image IDs.
- Caption text.
- Object category IDs and category names.
- Object-instance counts per image.
- Bounding boxes and other available instance attributes.

3.1.2. Dataset Size and Annotation Counts

Report the number of images in each split and the number of associated captions and object annotations.

For the knowledge base, the expected configuration is 10,000 images with five reference captions per image, corresponding to 50,000 captions.

Present a summary table containing:

- Number of images.
- Number of captions.
- Number of object annotations. Number of unique image IDs.
- Number of captions per image.

Use the actual outputs of the notebooks to complete this table.

3.1.3. Preliminary Data Quality Checks

Describe the checks used to identify:

- Missing image files.
- Corrupted images.
- Missing or empty captions.
- Unexpected numbers of captions per image.
- Duplicate image IDs.
- Duplicate images, detected through MD5 hashes.
- Duplicate captions associated with the same image.

Explain why these checks are important before constructing the knowledge base.

Suggested table: Data integrity checks and their results.

Check	Purpose	Result
Missing images	Identify unavailable image files	To be completed
Corrupted images	Detect unreadable image files	To be completed
Duplicate image IDs	Verify identifier uniqueness	To be completed

Duplicate images	Detect identical files	To be completed
Empty captions	Identify missing textual descriptions	To be completed
Incorrect caption counts	Check the expected five captions per image	To be completed

Do not report zero errors or other numerical results until they have been confirmed by executing the audit notebook.

3.2. Image Statistics and Visualization

3.2.1. Image Formats and Dimensions

Explain how image files are inspected using Pillow (PIL).

Analyze:

- Image formats.
- Image dimensions and resolution.
- Image color modes, such as RGB and grayscale. The distribution of image sizes.

Present the most frequent image dimensions and the number of images belonging to each color mode.

3.2.2. Visual Inspection of Sample Images

Show a grid of representative images selected from the knowledge base.

The project notebook already includes code for displaying a sample of nine images.

Use this figure to illustrate the diversity of scenes, object arrangements, and visual contexts present in the dataset.

3.2.3. Image Quality and Visual Diversity

Discuss the importance of verifying that images are readable and visually suitable for later processing.

If the necessary results are available, describe differences in image composition, dominant colors, and scene appearance.

The exploration notebook also contains an initial dominant-color analysis using a sample of 500 images. Present the results only if this analysis has been completed and interpreted.

Suggested figures:

- Examples of images from the dataset.
- Distribution of image dimensions.
- Distribution of image color modes.
- Optional visualization of dominant colors.

3.3. Object Category Distribution

3.3.1. Object Annotation Analysis

Explain that the instances_train2017.json file is used to identify the objects annotated in the knowledge-base images.

Each annotation is associated with an object category. These annotations make it possible to analyze which objects occur most frequently and how they are distributed across images.

The notebook excludes annotations marked as crowd instances from the principal categoryfrequency analysis.

3.3.2. Top 50 Most Frequent Object Categories

Describe how the frequency of each object category is calculated and how the 50 most frequent categories are selected.

Present the resulting table with:

- Category name.
- Number of object instances.
- Number of distinct images containing the category, where available.

Use a bar chart to visualize the 50 most frequent categories.

Discuss the main observations supported by your results, including which categories occur most frequently and whether the distribution is balanced or dominated by a smaller number of categories.

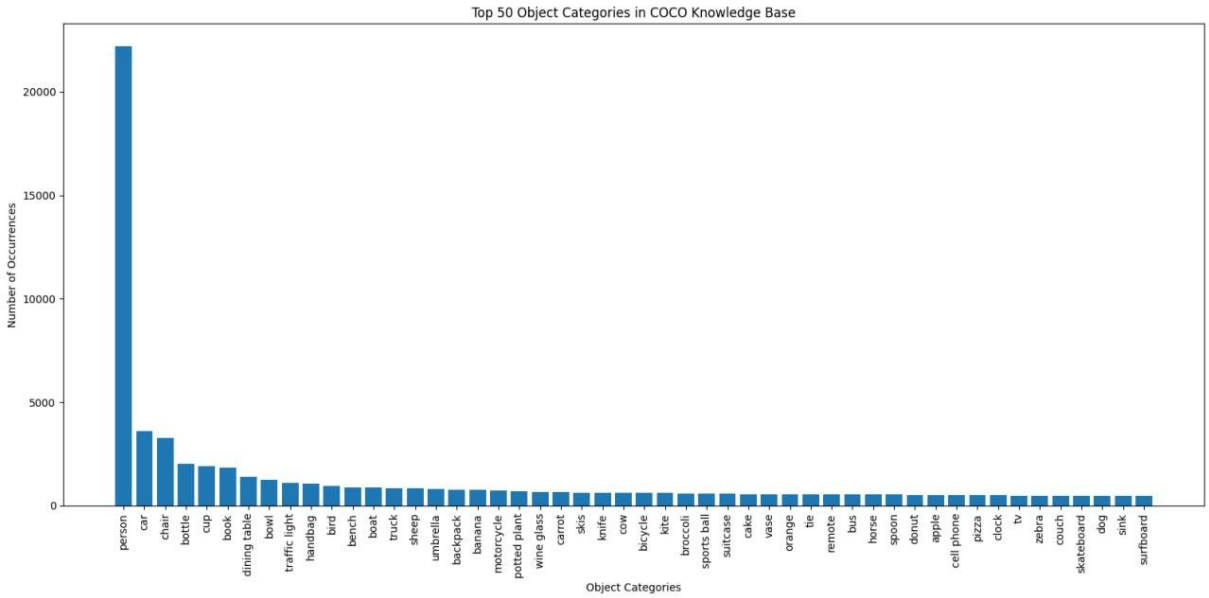


figure1-

3.3.3. Distribution of Categories Across Images

Analyze the number of distinct object categories appearing in each image.

Explain that this measure differs from the total number of objects: one image may contain many instances belonging to only a few categories.

This analysis helps characterize the semantic diversity of the scenes.

Suggested outputs:

- Table of the top 50 categories.
- Bar chart of the top 50 categories.
- Distribution of the number of distinct categories per image, if calculated.

Tableau1

	Category	Count			Category	Count
6	person	22182		64	Kite	630
8	car	3619		62	Broccoli	598
24	chair	3280		7	Sports ball	583
27	bottle	2040		47	Suitcase	570

0	cup	1905		10	Cake	562
11	book	1850		67	Vase	562
3	Dining table	1395		32	Orange	561
1	Bowl	1244		56	tie	558
41	Traffic light	1113		48	remote	553
44	handbag	1051		40	bus	541
46	bird	945		71	horse	539
13	bench	878		9	spoon	539
25	boat	869		42	donut	530
22	truck	857		31	apple	519
51	sheep	836		15	Cell phone	518
23	Umbrella	827		2	Pizza	503
43	Backpack	773		14	clock	499
30	banana	757		52	tv	494
5	motorcycle	726		18	zebra	492
63	Potted plant	704		29	couch	490
28	Wine glass	667		35	skateboard	489
68	Carrot	644		12	dog	471
58	skis	640		34	sink	467
37	Knife	636		26	surfboard	460

20	cow	634		39	Bicycle	632

Table1-

3.4. Caption Analysis

3.4.1. Reference Caption Extraction

Explain how the original captions are extracted from captions_train2017.json for the knowledgebase image IDs.

Each image is expected to have five human-written reference captions.

Describe the caption table used by the project, which associates each caption with its image_id.

3.4.2. Caption Length Distribution

Explain how caption length is measured by counting the words in each caption.

Present descriptive statistics, including:

- Mean caption length.
- Median caption length.
- Minimum and maximum length.
- Standard deviation.
- Caption length distribution.

The README reports an average caption length of approximately 10.49 words. This figure should be checked against the results produced by the current notebook before being reported as the final result.

Use a histogram to show the distribution of caption lengths.

Discuss whether captions tend to be short, descriptive, or variable in length.

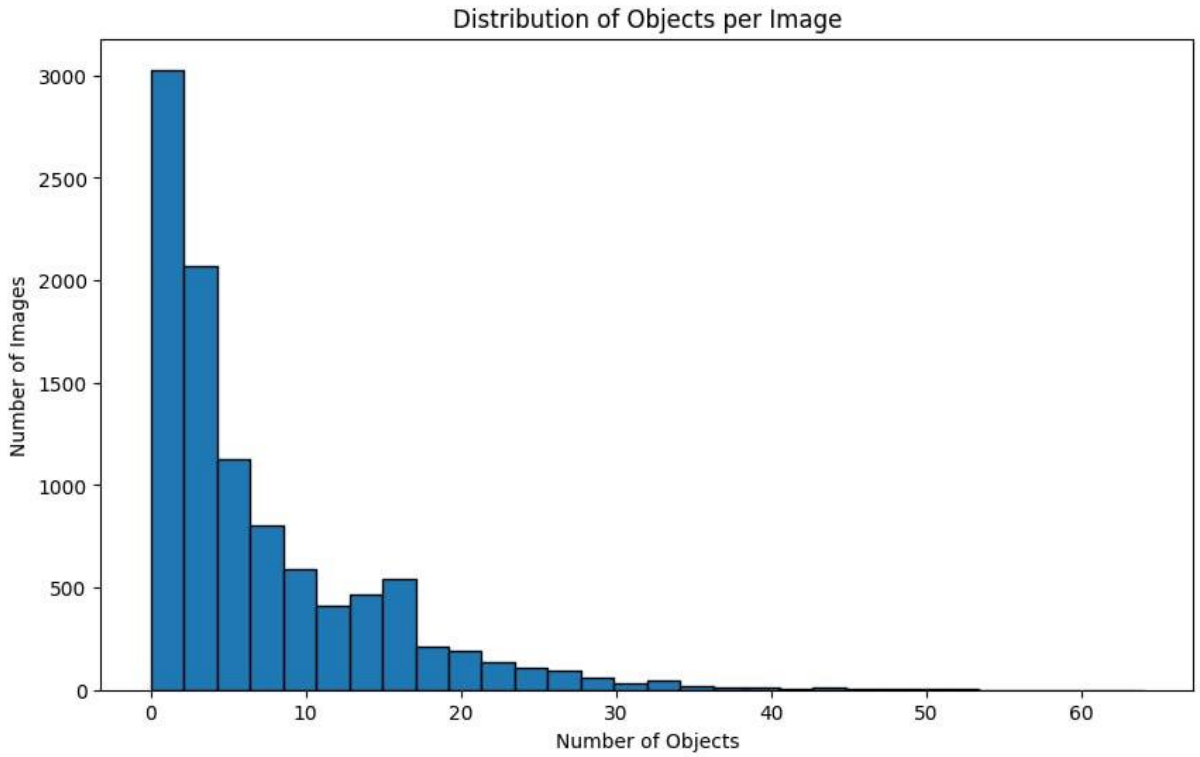


Figure2-

3.4.3. Vocabulary Analysis

Explain how words are extracted from the original captions after converting the text to lowercase for statistical analysis.

The current exploration notebook uses a regular expression to extract alphabetic words and calculates:

- Total word occurrences.
- Number of unique words. Most frequent words.

Present the 20 most frequent words in a table and bar chart.
Discuss what the vocabulary reveals about the descriptions in the dataset, including frequently mentioned objects and common descriptive expressions.

If stop words are removed for a supplementary analysis, clearly distinguish those results from the original frequency counts. **3.4.4. Caption Quality and Duplication**

Describe the checks for:

- Empty captions.

- Missing captions.
- Duplicate captions associated with the same image. Images with a caption count different from five.

Explain that repeated captions should be identified and documented rather than automatically deleted without considering their role in the original annotations.

3.4.5. Part-of-Speech Analysis

Describe a supplementary linguistic analysis of captions using part-of-speech (POS) tagging.

This analysis can examine the frequency of:

- Nouns, which identify objects and entities.
- Verbs, which describe actions and interactions.
- Adjectives, which describe attributes.
- Adverbs and other grammatical categories.

Explain how POS statistics could help characterize the linguistic information available in the captions.

Important: POS tagging is a useful extension of the analysis, but the current data_exploration.ipynb notebook does not implement it. Include this subsection only after adding the relevant code and obtaining actual results.

3.5. Scene Complexity and Diversity

3.5.1. Number of Objects per Image

Explain how the number of annotated objects is calculated for every knowledge-base image.

The notebook builds a table containing image_id and n_objects, then calculates descriptive statistics and plots a histogram.

Report the mean, median, minimum, maximum, and other useful statistics obtained from the executed notebook.
Use a histogram to illustrate the distribution of object counts.

Tableau

	Image_id	n_objects

0	101014	5
1	203463	1
2	462398	12
3	550444	15
4	36447	15

3.5.2. Classification of Scene Complexity

The project divides scenes into three groups:

- Simple scenes: no more than 3 annotated objects.
- Medium-complexity scenes: more than 3 and fewer than 10 annotated objects.
  Complex scenes: at least 10 annotated objects.

Report the number of images in each group and, if useful, the percentage of the knowledge base represented by each group.

Explain how this classification can support later comparisons of caption quality across scenes with different levels of complexity.

3.5.3. Object Co-occurrence

Describe the analysis of object categories that appear together in the same image.

The exploration notebook identifies pairs of distinct categories and counts the number of images in which each pair occurs.

Present the 20 most frequent category pairs in a table.

Discuss what these combinations reveal about common scenes and object interactions.

Tableau

	Object_1	Object_2	Images_together
0	person	car	725

1	person	chair	673
2	person	handbag	552
3	person	Dining table	514
4	chair	Dining table	461
5	person	cup	427
6	person	backpack	425
7	cup	Dining table	414
8	person	bottle	379
9	person	bench	376
10	person	Sports ball	372
11	person	truck	341
12	person	Cell phone	317
13	bowl	Dining table	307
14	car	truck	307
15	person	Tennis racket	306
16	person	tie	295
17	person	skateboard	280
18	person	umbrella	273
19	person	surfboard	270

3.5.4. Visual and Contextual Diversity

Discuss the different kinds of visual scenes present in the dataset, using the evidence available from image examples, object annotations, and category co-occurrence.

Possible aspects include:

- Variety of objects and object combinations.
- Simple versus complex scenes.
- Differences in scene composition. Differences in dominant colors.

The project's test-set design also aims to represent varied contexts and interactions.

Do not claim that indoor/outdoor, day/night, or other context labels have been quantitatively analyzed unless corresponding annotations or additional classification code are available.

3.5.5. Relationship Between Scene Complexity and Caption Length

As an additional analysis, compare caption lengths across simple, medium, and complex scenes.

For example, calculate the average caption length for images in each complexity group.

This could help determine whether images with more annotated objects tend to receive longer captions.

This analysis is a proposed extension; it should be included only after the relationship has been calculated from the dataset.

4. Data Pre-processing

4.1. Data Cleaning and Validation

4.1.1. Image Integrity Validation

Explain how the image preprocessing notebook checks the existence and readability of the knowledge-base images.

The implementation uses Image.open() and Image.verify() to detect files that cannot be read correctly.

Describe how missing or corrupted images are recorded and how these cases are handled before subsequent processing.

4.1.2. Duplicate Detection

Explain the distinction between:

- Duplicate image IDs, which can be detected by checking identifier uniqueness.
- Duplicate image files, which are checked through MD5 hashes.
  - Duplicate captions, which are identified in the caption table.

Report the actual findings from the audit.

4.1.3. Caption Validation

Describe the validation of raw and cleaned captions.

The project checks for missing values, empty captions, duplicate captions, and images with a number of captions different from five.

Explain why original captions are retained even when a cleaned version is created.

4.1.4. Validation Reports

Describe the audit information generated during preprocessing.

The caption preprocessing notebook creates caption_audit.json, which records the number of images, the total number of captions, missing-caption counts, and duplicate-caption counts.

Include a concise summary of the principal quality-control results.



Figure3-

4.2. Dataset Splitting and Leakage Prevention

4.2.1. Knowledge Base, Validation, and Test Sets

Explain the purpose of each split:

- The knowledge base contains the images and associated information available for retrieval.
- The validation set is used for development and parameter selection. The test set is reserved for final evaluation.

Describe how the image IDs are stored in splits/splits_all.json.

4.2.2. Split Verification

Explain the assertions implemented in splits_audit.ipynb to check:

- The expected number of image IDs in each split.
- The uniqueness of image IDs within each split.
- The relationship between validation and knowledge-base IDs.
- The disjointness of test IDs from knowledge-base and validation IDs.

Present a table summarizing the intended relationships between the splits.

4.2.3. Prevention of Information Leakage

Explain the project's anti-leakage rules:

- Test captions must not be included in the retrieval index.
- Test images must remain separate from the knowledge base.
- During validation retrieval, the query image must be excluded from its candidate results because validation images belong to the knowledge base. Test data must be used only for final evaluation.

Explain why these rules are necessary to ensure that the future RAG systems are evaluated fairly.

4.2.4. Preservation of the Split Configuration

Describe how the split configuration is stored and reused across notebooks.

The README states that the splits are frozen and must not be modified after finalization.

Mention the use of a fixed random seed for reproducibility.

Important: The split audit checks in the notebook should be consistent with the final splits_all.json file. Verify the actual relationships between the IDs before reporting that all assertions have passed.

4.3. Image Pre-processing

4.3.1. Loading and Inspecting Images

Describe how the images are loaded and inspected using Pillow.

Before applying transformations, the implementation records image formats, dimensions, and color modes.

4.3.2. EXIF Orientation Correction

Explain that some images contain EXIF metadata specifying how they should be displayed.

The project uses ImageOps.exif_transpose() to apply the appropriate orientation and avoid inconsistencies in image presentation.

4.3.3. Conversion to RGB

Explain why the project converts images to RGB.

Some images may be stored in grayscale or another color mode. Converting them to RGB provides a consistent three-channel representation for subsequent vision-model processing.

The implementation uses img.convert("RGB") and saves the processed images in data/processed/images/.

4.3.4. Image Resizing

Describe the resizing step implemented in preprocessing_images.ipynb.

The notebook uses Pillow's thumbnail() function with maximum dimensions of 384 × 512 pixels and the Lanczos resampling method.

Because thumbnail() preserves the original aspect ratio, the images are not forcibly stretched to identical dimensions.
The resized images are saved in data/processed/images_resized/.

Explain that the purpose of this step is to prepare image files for downstream processing while preserving their proportions.

4.3.5. Verification of Processed Images

Describe the checks performed after preprocessing:

- Number of output files.
- Image color modes after conversion.
- Dimensions of resized images. □ Processing errors.

Present a before-and-after summary showing which transformations were applied.

Do not state that all images were successfully processed until the corresponding counts have been verified.

4.3.6. Preparation for the Vision Encoder

The README identifies CLIP ViT-B/32 (openai/clip-vit-base-patch32) as the planned image encoder.

Explain that the encoder's preprocessing requirements must be respected when images are passed to the model.

The README refers to CLIPProcessor, which handles the model-specific resizing, cropping, and normalization.

Distinguish this model-specific processing from the separately saved images resized to a maximum of 384 × 512 pixels.

4.4. Caption Pre-processing-

4.4.1. Extraction of Original Captions

Explain how the original captions are retrieved from the COCO caption annotations using the image IDs in the knowledge base.

The notebook creates a table associating each caption with its image ID.

4.4.2. Creation of Clean Captions

Describe the `clean_caption()` function implemented in `preprocessing_captions.ipynb`. The current function:

- Converts text to lowercase.
- Replaces repeated whitespace with a single space.
- Removes characters other than word characters and whitespace. Removes leading and trailing whitespace.

Explain that these operations create a standardized textual representation for later retrieval experiments.

4.4.3. Preservation of Raw and Clean Versions

The project maintains two versions of each caption:

- `caption_raw`: the original COCO caption. `caption_clean`: the cleaned caption.

Explain why both are retained.

The original captions preserve the reference text for evaluation, whereas the cleaned captions can be used in retrieval-related processing.

The cleaned text should not automatically replace the original reference captions in the final evaluation.

Tableau

4.4.4. Caption Validation and Statistics

Describe the checks for:

- Missing original captions.
- Missing cleaned captions.
- Duplicate captions.
- Number of captions per image. Unique image IDs.

Explain how these checks help ensure that the caption table is consistent with the knowledgebase split.

4.4.5. Storage of Processed Captions

Describe the principal output files:

- `kb_captions_clean.csv`: the table containing the original and cleaned captions.
- `caption_audit.json`: a summary of the caption-quality checks.

Mention the columns and their meanings, as well as the relationship between captions and image IDs.

4.5.Metadata Extraction and Embedding Preparation

4.5.1. Object Metadata

Explain that object metadata are derived from the COCO instance annotations.

The project explores category IDs and names, object counts, category co-occurrence, and other available annotation attributes.

The README also identifies bounding boxes, supercategories, and object areas as metadata intended for preparation.

Clarify which metadata fields have actually been extracted and saved by the current implementation.

4.5.2. Caption and Image Metadata Organization

Describe how the project links image IDs with their captions and object information.

Explain why consistent identifiers are essential: they allow the future retrieval system to associate a retrieved representation with the corresponding image and textual information.

If a unified metadata table is created, document its columns and the relationship between its records and the original COCO annotations.

4.5.3. Planned Image and Text Embeddings

CLIP ViT-B/32, producing 512-dimensional image representations.

- Text encoder: Sentence-BERT `all-MiniLM-L6-v2`, producing 384-dimensional text representations.
- Similarity: cosine similarity using L2-normalized vectors.
- Vector index: FAISS `IndexFlatIP`, for exact inner-product similarity search.

Explain the general purpose of embeddings: they represent images or text as numerical vectors so that semantically similar items can be retrieved.

Clarify that the repository's current `embeddings.ipynb` mainly checks the PyTorch and Transformers environment and imports CLIP components. It does not yet implement the complete generation of embeddings or construction of the FAISS index.
Consequently, present embedding generation and index construction as planned or pending work unless these steps have been completed elsewhere.

4.5.4. Knowledge-Base Preparation

Explain how the previously described elements contribute to building the future retrieval corpus:

- Validated knowledge-base images.
- Associated reference captions.
- Object metadata.
- Image and text representations, once generated. A vector index, once constructed.

Explain that these components will provide the shared data foundation for the subsequent comparison of RAG techniques.

5. Results and Visualizations

5.1. Dataset Summary

Present a concise overview of the final dataset configuration.

Include:

- Number of images in the knowledge base.
- Number of validation and test images.
- Number of reference captions.
- Number of object annotations.
- Image-quality audit results.
- Number of images converted to RGB.
- Number of processed and resized images.

Use confirmed notebook outputs rather than expected values.

5.2. Object Distribution Results

Present the top 50 object categories and discuss the most important observations.

Include the category-frequency chart and scene-complexity statistics.

5.3. Caption Statistics Results

Present caption-length statistics, vocabulary size, frequent-word analysis, and caption-quality checks.
Include the relevant tables and figures.

If POS tagging has been implemented, summarize the grammatical-category distribution here.

5.4. Scene Diversity Results

Present the distribution of objects per image, the simple/medium/complex scene counts, and the most frequent object co-occurrences.

Relate these observations to the diversity of scenes that the future captioning systems will encounter.

5.5. Pre-processing Results

Summarize the transformations and validation operations performed.

A useful table is:

Data component	Operation	Output
Images	EXIF orientation correction	Corrected RGB images
Images	RGB conversion	`data/processed/images/`
Images	Resizing with preserved aspect ratio	`data/processed/images_resized/`
Captions	Original caption extraction	Caption table
Captions	Text cleaning	`caption_clean`
Captions	Quality audit	`caption_audit.json`
Object annotations	Category and scene analysis	Exploration tables
Dataset splits	ID verification	`splits/splits_all.json`

5.6. Final Data Artifacts

Describe the files produced during Phase 1 and explain how each will be used later.

Distinguish between:

- Files already generated and verified.
- Files generated by notebooks but not yet confirmed.
- Planned outputs, such as the final embeddings and FAISS index, that still require implementation or verification.

This distinction makes the report transparent and reproducible.

6. Conclusion

Summarize the work completed during data exploration and preprocessing.

Highlight the importance of:

- Understanding the dataset's object and caption distributions.
- Validating the integrity of the images and captions.
- Standardizing image formats and orientation.
- Preserving original reference captions.
- Maintaining reliable dataset splits and preventing information leakage. Preparing metadata for future retrieval experiments.

Conclude by explaining that the quality and consistency of these data artifacts are essential for a fair comparison of RAG techniques in the next phases.

Briefly identify any remaining tasks, such as completing POS analysis, verifying all audit results, finalizing metadata extraction, and generating embeddings and the vector index if they are still pending.